

Lecture 1 Live Problems

Welcome! What is this class about?

Points toward lab manual pages: **ST1, ST2, ST3, ST4** (section L1 S1)

1. Turkeys and coyotes. Declare variables to represent the populations of turkeys and coyotes on campus.

– Let T be _____

– Let C be _____

Basic assumptions: the turkey population grows quickly without coyotes. But coyotes eat turkeys. When coyotes eat turkeys, the turkey population goes down, and the coyote population goes up because they are well-fed and healthy. Without eating turkeys, the coyotes die off.

2. Draw a time-series plot of the turkey and coyote populations, assuming we start with few coyotes and many turkeys.



3. Draw the corresponding *trajectory in state space*.

